SIMULATING NITRIC OXIDE DIFFUSION WITHIN MOF-BASED POLYMERS USED TO PREVENT CATHETER RELATED THROMBOSIS AND INFECTION

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INTRODUCTION

Infection and thrombosis are persistent risks associated with catheterization. Nitric oxide (NO), a gaseous signalling molecule with antimicrobial and vasodilatory properties, has demonstrated efficacy in mitigating these complications [1]. Metal–Organic Frameworks (MOFs), due to their high surface area and porosity, are promising carriers for NO storage and controlled release via water-triggered exchange (Fig. 1A) [2].

In this work, NO is present in the MOF–polymer composite in two forms: physisorbed NO, which diffuses out under a concentration gradient, and chemisorbed NO, which is released upon moisture exposure during catheter use. This study refines a multiphase finite element (FE) simulation of NO release using experimentally validated material diffusion parameters to inform catheter design for controlled release (Fig1C).

MATERIALS AND METHODS

Tubular specimens containing 5 wt.% Ni-based MOFs were extruded in dimensions matching cardiovascular catheters. NO and H_2O diffusion properties were experimentally measured using standardised ISO/DIN gas diffusivity methods by a permeation testing company (Mecadi GmbH, Bexbach, Germany).

A parametric finite element model was constructed using FEBio 4.9 with quarter-symmetry geometry, incorporating both material and environmental variables. Moisture-triggered release reactions were embedded into the material definition to simulate dynamic NO release and outward diffusion. The model was benchmarked against experimental NO release profiles obtained project partners.

RESULTS & DISCUSSION

Simulations revealed distinct NO release behaviours: chemisorbed NO showed a sustained, gradual release, while physisorbed NO exhibited an initial burst (Fig. 1B). Simulated and experimental data matched well in terms of total NO released over 4 hours at 95% relative humidity, but the model failed to capture the sharp initial rise and gradual tailing seen in the experimental release curve (Fig. 1D). These differences likely result from restricted diffusion pathways in semi-crystalline Pebax or temporary trapping within rigid amorphous regions [3]. The results highlight the critical roles of water uptake, NO binding type, and the material's microstructure in shaping the release profile.

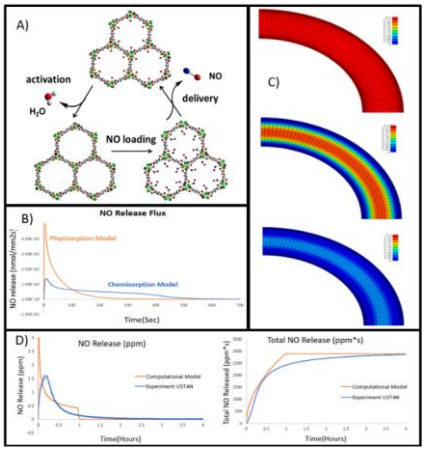

Figure 1 (A) Cycle of loading and releasing NO from MOFs. (B) Release profiles of NO bounded with polymer by Physisorption and Chemisorption. (C) Simulation of NO release through polymer. (D)NO release profiles compared to experiments

CONCLUSION

Results demonstrate that a multiphase FE model can predict realistic release times, although struggles to match the shape of the release profile. Future work will incorporate more accurate experimental parameters that reflect the polymer's heterogeneous structure to improve model fidelity. With sufficient accuracy, the model could support computational pre-screening of catheter designs, reducing experimental iterations while ensuring therapeutic efficacy.

ACKNOWLEDGEMENTS

This work was supported by the Engineering and Physical Sciences Research Council [EP/X014436/1]. Special thanks to our project collaborators for supplying material samples & experimental data.

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